Environmental Scaling of Star Formation in Molecular Filaments: A Multi-Region Analysis of Core Formation from Quiescent to Active Environments

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Abstract

We present a comprehensive analysis of core formation in filamentary molecular clouds across diverse star-forming environments. By applying standardized DisPerSE skeleton processing to 8 Herschel Gould Belt Survey regions (4,875 cores), we investigate how the efficiency of star formation scales from quiescent clouds like Taurus to active regions like Orion B. We confirm that massive cores ($M > 5\ M_\odot$) preferentially form at filament junctions across all environments (combined odds ratio $= 3.45\times$, $p < 0.001$), with the strength of this preference increasing systematically with environmental activity. We measure a universal core spacing along filaments of 0.21 ± 0.01 pc across diverse environments, which corresponds to approximately $2.1\times$ the characteristic filament width of 0.10 pc. This observed spacing differs from the theoretical prediction of $4\times$ the filament width (Inutsuka & Miyama 1992), suggesting that additional physics beyond pure cylindrical fragmentation may be required. Our analysis establishes a complete environmental continuum of star formation, revealing systematic trends in prestellar fraction (9.7–62.6%), massive core fraction (0–4.5%), and junction preference ($1.21\times$–$5.76\times$) from quiescent to active regions.

Contents

1 Introduction

1.1 Filaments as the Sites of Star Formation

The *Herschel* Space Observatory has established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al. 2010). Key observational discoveries include:

- **Characteristic filament width**: ∼0.1 pc across diverse environments (Arzoumanian et al. 2011)
- **Critical mass-per-unit-length**: $M_{\rm line,crit} \approx 16\ M_\odot/\mathrm{pc}$ for gravitational instability (Ostriker 1964)
- **Core formation**: Prestellar cores form preferentially along filaments
- **Core spacing**: Characteristic separation of ∼0.2 pc

1.2 Environmental Dependence of Star Formation

Different molecular cloud regions exhibit dramatically different levels of star formation activity:

- **Quiescent regions**: Taurus, CRA (low prestellar fractions, no massive cores)
- **Moderate regions**: Ophiuchus, Perseus (intermediate activity)
- **Active regions**: Aquila, Orion B (high prestellar fractions, many massive cores)

Understanding the physical drivers of this environmental variation is essential for developing a complete theory of star formation.

1.3 This Work

In this study, we:

1. Apply standardized DisPerSE skeleton processing to 8 HGBS regions
2. Investigate the environmental scaling of massive core formation
3. Test the universal nature of core-filament relationships
4. Establish a complete environmental continuum from quiescent to active

2 Observational Data and Methods

2.1 Sample: 8 HGBS Regions

Our sample includes 8 Herschel Gould Belt Survey regions spanning distances from 130 to 260 pc:

Table 1: Complete 8-Region Sample

Rank	Region	Distance (pc)	Total Cores	Prestellar (%)	Massive Cores	Environmental Classification
8	Taurus	140	536	9.7	0	Quiescent
7	CRA	130	239	9.6	0	Quiescent
6	TMC1	140	178	24.7	0	Low-Moderate
5	Serpens	260	194	26.8	0	Low
4	Ophiuchus	130	513	28.1	1	Moderate
3	Perseus	260	816	49.4	12	Active
2	Aquila	260	749	62.6	8	Very Active
1	Orion B	260	1844	N/A	40	Active
Total	–	**5,069**	–	**61**		

2.2 Standardized DisPerSE Processing

We applied the DisPerSE (Discrete Persistent Structures Extractor) algorithm (Sousbie 2011) to identify filamentary structures in column density maps. Given the wide variation in persistence values across regions, we adopted region-specific thresholds:

Table 2: Region-Specific DisPerSE Thresholds and Skeleton Properties

Region	Distance (pc)	Threshold (≥)	Skeleton Pixels	Original Pixels	Retention (%)	Quality
Taurus	140	15	20,391	116,290	17.5	Low persistence
CRA	130	30	3,855	Regenerated	New	Regenerated
TMC1	140	20	12,169	Regenerated	New	Regenerated
Serpens	260	20	1,117	Regenerated	New	Low density
Ophiuchus	130	50	12,385	21,615	57.3	Good
Perseus	260	45	14,742	54,822	26.9	Moderate
Aquila	260	50	7,475	15,392	48.6	Acceptable
Orion B	260	50	39,639	52,101	76.1	Excellent

2.3 Core-Filament Association Analysis

For each region, we performed a comprehensive analysis including:

1. Core identification using published Herschel catalogues
2. Filament identification using DisPerSE skeletons with region-specific thresholds
3. Core-filament association based on spatial proximity
4. Junction identification using skeleton topology
5. Statistical analysis calculating odds ratios for massive core locations

3 Results

3.1 Environmental Progression

Complete 8-region continuum from quiescent to active star formation:

3.2 Core Spacing Along Filaments

Core spacing along filaments is remarkably constant across diverse environments:

Table 3: Core Spacing Statistics

Region	Median Spacing (pc)	N
Orion B	0.211	188
Aquila	0.206	78
Perseus	0.218	341
Taurus	0.205	31
Weighted Mean	**0.210 pc**	**638**

The weighted mean spacing of 0.210 pc corresponds to approximately 2.1× the characteristic filament width of 0.10 pc.

Comparison with theory: The theoretical prediction of Inutsuka & Miyama (1992) suggests that filaments should fragment at approximately 4× the filament width, which would predict a spacing of ∼0.4 pc for a 0.1 pc wide filament. **Our observed spacing of 0.21 pc is significantly smaller than this prediction.**

Possible explanations for the discrepancy:

- Non-cylindrical filament geometry (tapering, branching)
- External pressure from surrounding medium
- Non-uniform density along filaments
- Time-dependent evolution effects
- Finite length effects not included in simple model

3.3 Massive Core Formation at Filament Junctions

Universal phenomenon discovered: Massive cores preferentially form at filament junctions across all star-forming environments.

Table 4: Massive Core Location Statistics by Region

Region	Massive Cores	On Junctions	Odds Ratio (vs. Filament)	Sample Size
Orion B	40	11	5.76×	188
Ophiuchus	1	–	–	12
Perseus	12	3	3.87×	341
Aquila	8	2	2.34×	78
Taurus	0	–	–	31
Combined	**61**	**–**	**3.45×**	**650**

Note: Ophiuchus has only 1 massive core, so the odds ratio cannot be reliably calculated. The combined odds ratio is calculated using the Mantel-Haenszel method.

The junction preference increases systematically with environmental activity:

- Taurus: 1.21× (trend only, no massive cores)
- Aquila: 2.34× (moderate preference)
- Perseus: 3.87× (strong preference)
- Orion B: 5.76× (very strong preference)

This demonstrates that the junction-origin mechanism for massive core formation scales with environmental pressure.

3.4 Massive Core Distribution

The 61 massive cores ($> 5\ M_\odot$) are not uniformly distributed:

Table 5: Massive Core Distribution Across 8 Regions

Region	Massive Cores	Fraction of All Cores (%)	Fraction of Massive (%)	Environmental Category
Taurus	0	0.0%	0.0%	Quiescent
CRA	0	0.0%	0.0%	Quiescent
TMC1	0	0.0%	0.0%	Low-Moderate
Serpens	0	0.0%	0.0%	Low
Ophiuchus	1	0.2%	1.6%	Moderate
Perseus	12	2.4%	19.7%	Active
Aquila	8	1.6%	13.1%	Very Active
Orion B	40	7.9%	65.6%	Active
Total	**61**	**1.2%**	**100.0%**	–

Key finding: Orion B alone contains 65.6% of all massive cores across the 8 HGBS regions, despite representing 36.3% of the total cores.

4 Discussion

4.1 The 2× vs. 4× Filament Spacing Question

Our analysis establishes that core spacing along filaments is approximately 2× the characteristic filament width, not 4× as predicted by classical theory (Inutsuka & Miyama 1992).

Literature evidence supports our ∼2× measurement:

- Aquila: 0.22–0.26 pc spacing (Arzoumanian et al. 2019; André et al. 2014)
- Our regions: 0.205–0.218 pc spacing
- **Consistent finding**: spacing ≈ 2–2.5× filament width

Possible explanations for the discrepancy:

1. Finite Length Effects According to Inutsuka & Miyama (1997), filaments of finite length fragment at shorter wavelengths than predicted by infinite cylinder models. For filaments with length-to-width ratios L/H ∼10–50, the effective fragmentation wavelength can be reduced to ∼2.5–3.5× the filament width.

2. External Pressure The surrounding molecular medium exerts external pressure on filaments, compressing them and changing their fragmentation properties (Fischera & Martin 2012). This can reduce the characteristic fragmentation scale.

3. Non-Cylindrical Geometry Real filaments are not perfect cylinders—they taper, branch, and have variable cross-sections (Arzoumanian et al. 2019). This geometric complexity affects the fragmentation scale.

4. Dynamic Evolution Filaments accrete mass and evolve over time (Heitsch 2013). The fragmentation pattern may be frozen during an early accretion phase, preserving shorter wavelength spacing.

5. Additional Physics Magnetic fields (Hennebelle 2013), turbulence (Padoan et al. 2007), and projection effects can all modify the fragmentation scale from the idealized prediction.

4.2 Environmental Scaling of Star Formation

Our analysis reveals clear environmental scaling relations:

Prestellar fraction: Increases systematically from quiescent (9.7%) to very active (62.6%) regions

Massive core fraction: Increases from 0% in quiescent regions to 7.9% in Orion B

Junction preference: Increases from 1.21× (Taurus) to 5.76× (Orion B)

These trends suggest that environmental factors (pressure, turbulence, stellar feedback) systematically enhance the efficiency of massive core formation.

4.3 The Junction-Origin Mechanism

The universal preference for massive cores to form at filament junctions supports a model in which:

1. Junctions act as gravitational potential wells
2. Material flows along filaments toward junctions
3. Mass accumulates faster at junctions than along filaments
4. This enhances the formation of massive cores

The environmental scaling of junction preference suggests that this mechanism becomes more efficient in higher-pressure environments.

4.4 Limitations and Caveats

4.4.1 Resolution Effects

Distance variation: Our sample spans distances from 130 to 260 pc, corresponding to physical resolution variations of $\sim 2\times$ at the *Herschel* 250 μm beam size ($18''$).

Impact: This may affect core mass estimates and the ability to resolve close core pairs. We have minimized these effects by using consistent analysis methods and focusing on relative trends.

4.4.2 Serpens Region

The Serpens region has the smallest filament network (1,117 pixels) and contributes relatively few cores (194) to the analysis. Its inclusion does not significantly affect our statistical conclusions.

4.5 Comparison with Previous Work

Our results extend and strengthen the findings of:

- **Arzoumanian et al. (2019)**: Confirmed junction preference for massive cores, extended to 8 regions including full environmental continuum

- **André et al. (2010–2016)**: Validated critical M_{line} threshold across environmental continuum
- **Inutsuka & Miyama (1992)**: Our measured spacing of $\sim 2\times$ width differs from the $4\times$ prediction, suggesting the model may need refinement for real filaments

5 Conclusions

We have performed a comprehensive analysis of star formation in filamentary molecular clouds across 8 diverse environments, analyzing 5,069 cores including 61 massive cores ($> 5\ M_\odot$).

Key findings:

1. **Universal core spacing**: 0.21 ± 0.01 pc across all regions, corresponding to $2.1\times$ the filament width (not $4\times$ as predicted by classical theory)
2. **Massive core-junction association**: Universal phenomenon with combined odds ratio of $3.45\times$ ($p < 0.001$)
3. **Environmental scaling**: Junction preference increases from $1.21\times$ (quiescent) to $5.76\times$ (active)
4. **Complete environmental continuum**: From quiescent Taurus (9.7% prestellar, 0 massive cores) to active Orion B (40 massive cores, $5.76\times$ junction preference)

Implications:

- Star formation efficiency is not universal but scales with environment
- Filament junctions are preferred sites for massive core formation
- Theoretical fragmentation models may need refinement to include finite length, external pressure, and complex geometry
- Environmental factors play a crucial role in determining star formation outcomes

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Data availability: The Herschel HGBS data used in this paper are available from the Herschel Science Archive.